THE FREE-ALGEBRA FUNCTOR IN GROUPOIDS IS THE ACTION GROUPOID

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Source reference. Original equation 05YN: sources/stacks-criteria/pinned-original-source.tex, source lines 2413–2413. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 05YM: sources/stacks-criteria/pinned-original-source.tex, source lines 2391–2391. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *The functor in groupoids FA_d defined in (05YN) is isomorphic (!) to the functor in groupoids which associates to a scheme T the category with*

1. *set of objects is $X(T)$,*
2. *set of morphisms is $G(T) \times X(T)$,*
3. *$s : G(T) \times X(T) \to X(T)$ is the projection map,*
4. *$t : G(T) \times X(T) \to X(T)$ is $a(T)$, and*
5. *composition $G(T) \times X(T) \times_{s,X(T),t} G(T) \times X(T) \to G(T) \times X(T)$ is given by $((g, m), (g', m')) \mapsto (gg', m')$.*

Proof. We have seen the rule on objects in (05YM). We have also seen above that $g \in G(T)$ can be viewed as a morphism from m to $a(g, m)$ for any free d-dimensional algebra m. Conversely, any morphism $m \to m'$ is given by an invertible linear map φ which corresponds to an element $g \in G(T)$ such that $m' = a(g, m)$. □

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